CareerPilotAI

Updated Project Goals, Decisions & Implementation Status

Project direction: Build a professional, AI-first career platform that helps users understand their resumes, explore relevant jobs, prepare for interviews, and plan career development.

Document purpose

This document replaces the earlier goals-only snapshot with a working reference that distinguishes completed work, existing functionality that still needs verification, and future improvements. Status labels reflect the project information currently available; they are not a substitute for testing the latest repository.

Updated: 20 October 2026

Frontend constraint: HTML5, CSS, Vanilla JavaScript (ES6), and Tailwind CSS via CDN. No React, Vue, Angular, TypeScript, Bootstrap, or npm-based frontend build system.

1. Project Vision

CareerPilotAI is an AI-first career guidance platform designed as a startup-style MVP. It combines resume understanding, ATS-oriented feedback, job recommendations, interview practice, and personalized career guidance.

Core user journey

- Create an account and authenticate securely.
- Upload a resume and extract its content.
- Review resume analysis, ATS-oriented feedback, skills, strengths, weaknesses, and suggestions.
- Explore job recommendations using semantic similarity and AI-assisted ranking where configured.
- Use the career mentor, mock interview, and career roadmap features to prepare for career opportunities.
- Use future enhancements to connect identified skill gaps to learning plans and tailored application materials.

2. Current Implementation Status

The following status is based on project details and previous implementation notes available for this update. Features marked “implemented; retest” should be checked against the latest local code before being presented as fully verified.

Area	Current status	Known implementation / note
Backend foundation	**Completed**	FastAPI backend and modular feature structure are reported as implemented. Preserve existing API contracts during frontend work.
Authentication	**Implemented; retest**	JWT-based authentication and user-related endpoints are part of the project. Email verification remains a future enhancement.
Resume upload and parsing	**Implemented; retest**	Resume management and extraction using document parsing/OCR tooling have been described in the project.
Resume analysis / ATS	**Implemented; retest**	ATS score, skill extraction, strengths/weaknesses, and AI suggestions are existing capabilities described in project notes.
Job recommendations	**Implemented; retest**	Semantic similarity and AI-assisted ranking are described. A clear user-facing explanation of each match needs verification.
AI career mentor	**Implemented; retest**	Career guidance/chat and conversation history are described in the existing module set.
Mock interviews	**Implemented; retest**	AI-generated questions, answer evaluation, feedback, scoring, final report, and PDF export have been described. Cross-interview trend analytics are not confirmed.
Course recommendations / roadmap	**Implemented or in project scope; verify**	Course and career roadmap modules are part of the architecture. Confirm current UI, data coverage, and end-to-end behavior before claiming completion.
Frontend redesign	**Planned**	Redesign the existing frontend module by module with a minimal professional visual system and dark/light theme toggle.

Important: “Implemented; retest” means the feature is reported as existing, not that it has been freshly tested in this update. Avoid treating an item as complete until its current UI and API behavior have been checked.

3. Technology Decisions

Frontend

- **HTML5, CSS, Vanilla JavaScript (ES6), and Tailwind CSS via CDN.**
- Font Awesome via CDN may be used for icons where already present.
- No React, Vue, Angular, TypeScript, Bootstrap, or npm-based frontend build pipeline.
- The frontend remains an API client; do not move business logic from the backend into the UI or alter API contracts just to support styling.

Backend and data

- FastAPI for REST APIs and application services.
- SQLAlchemy for database access; use the current configured database and migrations rather than assuming the database engine from older planning documents.
- Pydantic for request/response validation and JWT for authentication.
- Keep secrets in environment variables and never commit `.env` files or API keys.

AI and document processing

- AI provider integration is configurable through the backend settings; current project notes mention OpenRouter-compatible endpoints and APINex testing.
- Keep the embedding/similarity provider separate from the chat/generation provider unless a deliberate change is tested.
- Resume processing has used OCR/document extraction tooling such as EasyOCR and pdfplumber; verify actual installed dependencies before changing them.
- Use semantic matching where supported and generate explanations with AI when appropriate; avoid duplicate hardcoded logic.

4. Architecture & Development Principles

- Maintain a modular backend and a separate frontend.
- Preserve existing routes, request/response schemas, authentication behavior, and database schema during visual redesign.
- Keep functions focused and avoid duplicated logic.
- Prefer small, reversible changes; test each module before proceeding.
- Use Git commits with descriptive messages and keep the main branch stable.
- Do not introduce a new framework or dependency solely for visual polish.

5. Frontend Redesign Plan

Chosen direction: minimal professional design, with both dark and light themes controlled by a theme toggle. Redesign the whole frontend one module at a time.

Phase 1 — Audit and protect existing behavior

- Inventory all current HTML pages, CSS files, JavaScript modules, and API calls.
- Record authentication flow, navigation, form behavior, upload flow, and API contracts before edits.
- Use the latest local frontend as the source of truth; do not assume the public repository contains the latest UI.

Phase 2 — Shared design foundation

- Define CSS variables for light/dark surface, text, borders, muted text, and accent colors.
- Standardize typography, spacing, buttons, form controls, cards, alerts, loading indicators, and empty states.
- Implement a theme toggle with local preference persistence and a sensible initial theme.
- Check contrast, keyboard focus, reduced motion, and mobile layouts.

Phase 3 — Module-by-module redesign

- Start with the dashboard to establish the design system.
- Continue with authentication, resume management, resume analysis/ATS, jobs, career mentor, mock interviews, courses, and roadmap according to the actual existing navigation.
- Change presentation and interaction feedback without changing backend endpoints or data contracts.

Phase 4 — Verification after every module

- Test in both themes and at desktop/mobile widths.
- Verify sign-in/sign-out, token handling, page navigation, uploads, API success/failure, and loading states.
- Check browser console and network requests for errors.
- Proceed to the next module only after the current module passes its checks.

Smallest safe first change

Create the shared theme variables and theme toggle first, then apply the foundation to the dashboard. Avoid a full rewrite, broad CSS replacement, or backend/database changes.

6. Next Improvements Roadmap

A. Small improvements — do these first

- Improve loading, success, empty, and error states in existing pages.
- Add clear form validation and prevent accidental duplicate submissions.
- Improve responsive layouts, navigation consistency, button feedback, and accessibility.
- Review configuration and security handling without exposing secrets or changing behavior unnecessarily.

B. Product improvements — implement after the small fixes

- **Job Match Explanation:** show why a job matches a resume, including relevant strengths, matching skills, and missing requirements. First verify what the current job results already display.
- **Skill Gap Analyzer:** compare a target job's requirements with extracted resume skills and recommend relevant courses or learning resources. First verify whether current resume analysis already supports a job-specific gap workflow.
- **AI Resume Improvement Editor:** propose clearer, evidence-based bullet points and show before/after versions tailored to a target role. Do not invent experience, qualifications, or achievements.
- **Interview Performance Analytics:** deferred for now. Existing interview evaluation and reports are already described; historical score trends and recurring weaknesses across multiple interviews are not confirmed.

C. Future enhancements

- Email verification.
- Voice-to-text interview responses using browser speech recognition where supported, with a text fallback.
- Timed mock interviews with countdowns and per-question/session time limits.
- AI-assisted job application drafts based on the resume and target job description, requiring user review before use.
- Weakness-focused application optimization that addresses genuine skill gaps while accurately presenting relevant strengths and experience.
- Live job listings from LinkedIn, Naukri, and other platforms only through authorized APIs, permitted feeds, or other compliant access methods.
- Resume version comparison, company-specific interview preparation, multi-language support, notifications, deployment, and further analytics as later priorities.

7. Completion Rules

- A feature is “completed” only after its actual implementation has been inspected and its key user flow tested.
- A visual redesign is successful only if existing functionality continues to work in both themes.
- New features must reuse existing modules and data where appropriate rather than duplicating functionality.
- Keep scope controlled: one module, one change set, one test pass at a time.

Project goal

Deliver a polished, reliable AI career platform with a professional user experience, explainable recommendations, useful resume feedback, and realistic interview practice—while preserving the existing backend architecture and keeping changes incremental.

Status note: This document consolidates the provided project-goals PDF and previously discussed CareerPilotAI implementation details. It does not claim that the latest GitHub repository was freshly audited or that every listed existing feature has been retested.